LINEAR ELECTRONICS
ASSIGNMENT 1

1. A transformer-coupled Class A amplifier is designed to deliver a maximum average AC power output of $P_{AC(\max)} = 5$ Wto an $R_L = 8$ speaker. The amplifier is powered by a DC supply $V_{CC} = 18$ V. The output transformer is designed for optimum impedance matching, i.e. the load reflected to the primary winding is perfectly matched to the transistor.

   Required:

   - Part A: Determine the maximum DC power input $P_{DC(\max)}$required for the amplifier (use the appropriate Class A transformer-coupled efficiency assumption).
   - Part B: Calculate the quiescent collector current I_{CQ}required to achieve this output power.
   - Part C: Calculate the maximum power dissipation $P_{D(\max)}$required of the transistor, and explain why this is a critical specification for Class A amplifier design.

2. Onboard a ship, the engine room communication system uses a small Class A audio amplifier to drive a speaker used for local alarms and voice announcements.
   The amplifier uses a single-transistor common-emitter configuration powered from a 15 V DC supply.
   The design aims for low distortion and reliable operation rather than high efficiency — a common requirement in marine control circuits where signal clarity is critical.

   The amplifier is biased such that the quiescent (no-signal) collector current is:

   $$I_{CQ} = 50 \text{ mA}$$

   and the collector resistor is:

   $$R_C = 150\ \Omega$$

   Tasks

   Part A:
   Calculate the Quiescent Collector–Emitter Voltage $(V_{CEQ})$and the DC Power Consumed $(P_{DC})$by the amplifier under no-signal conditions.

   Part B:
   Determine the Quiescent Power Dissipation in the transistor $(P_{D(Q)})$.

Part C:
If the amplifier is driven to its maximum symmetrical output swing (without clipping), determine:

- the maximum theoretical AC power output $(P_{AC(max)})$, and
- the maximum efficiency $(\eta_{max})$of this Class A amplifier.

3. A vessel's onboard entertainment and alarm system uses a Class B push–pull audio amplifier to drive a loudspeaker in the mess and control room. The amplifier is powered from the ship's power conversion system, which provides a split supply for the amplifier rails. Maintenance engineers must verify amplifier ratings and thermal margins before commissioning.

   Scenario:
   The amplifier is powered by a split DC supply of ± 25 V. It drives a single speaker of impedance $R_L = 16$. Under test conditions the amplifier must be capable of delivering the specified maximum clean output; during acceptance testing the measured peak output voltage across the speaker is $V_{\text{peak}} = 22$ V.

   Assumptions: ideal complementary push–pull output stage, sinusoidal steady-state output, no additional series impedance unless stated.

   Calculate and report values

   Part A: Calculate the maximum AC power output delivered to the speaker, $P_{AC(\max)}$, based on the given peak voltage.

   Part B: Calculate the total DC power consumed from the ± 25 Vrails, P_{DC}, under these maximum output conditions.

   Part C: Determine the collector efficiency ηat maximum output and compute the total power dissipated as heat by both output transistors, $P_{D(\text{Total})}$. Comment briefly on the thermal implications for heatsinking in the shipboard installation.

4. The ship's bridge communications and alert system use a small transformer-coupled Class A audio amplifier to drive a cockpit/bridge speaker. For safety and audibility, it must reliably produce a continuous, low-distortion audio tone for alarms. The electrical engineering team specifies the amplifier so it can deliver a known maximum usable

audio power into the standard ship speaker while keeping transistor dissipation and heatsinking within safe limits.

System specification:

- Load (speaker): $R_L = 8\ \Omega$
- Required maximum average AC power delivered to the speaker:

$$P_{AC(\max)} = 5\ \text{W}$$

- DC supply for the amplifier:

$$V_{CC} = 18\ \text{V}$$

- Output coupling: transformer-coupled and optimally matched — i.e. the load as seen at the primary is the impedance required for ideal Class A operation (the transformer is designed so the reflected load is the correct impedance for the transistor stage).

Tasks

Part A
Determine the maximum DC power the amplifier must draw from the supply to support the required $P_{AC(\max)} = 5$W output, using the appropriate theoretical efficiency assumption for a transformer-coupled Class A stage. (State any efficiency assumption used.)

Part B
Calculate the quiescent collector current $I_{CQ}$the transistor must be biased at so that, with the chosen transformer and load, the amplifier can produce the specified 5 Wmaximum average AC output without clipping. (Show how I_{CQ}relates to the required peak output current and your chosen Class A operating point.)

Part C
Calculate the maximum power the transistor must be able to dissipate (worst-case dissipation) at this operating point. Explain why this transistor dissipation rating is a critical specification in Class A design for shipboard equipment (consider continuous operation, safety margins, and thermal management). Include the governing equation(s) used and state any assumptions (for example, about symmetric swing, ideal transformer coupling, or use of peak vs average quantities).